

Node-Pruning Comparison

Normal node pruning at two operating points versus the final position-aware checkpoint

Circuit sizes

Circuit	High-level edges active/full	Heads	Attn neurons closed	MLPs	MLP hidden closed	MLP output closed	Model proxy closed
Normal - KL matched (ep. 80)	2,008/11,754 (82.9% closed)	51/144	71.5%	12/12	83.5%	66.2%	56.2%
Normal - epoch matched (final)	1,703/11,754 (85.5% closed)	48/144	73.8%	10/12	88.3%	74.6%	58.3%
Position-aware - final	1,276/109,722 (98.8% closed)	75/1,296	95.4%	25/108	95.6%	91.2%	49.8%

Performance on the shared 500-example test set

Circuit	Accuracy	Exact	Pairwise	KL	Logit diff	Faithfulness
Normal - KL matched (ep. 80)	98.4%	98.4%	100.0%	0.154	6.414	1.503
Normal - epoch matched (final)	97.2%	97.2%	100.0%	0.194	6.525	1.529
Position-aware - final	98.8%	98.8%	100.0%	0.143	6.979	1.635

All rows use hard, hierarchy-finalized gates. Epoch 80 is the closest tested normal-node KL match to the positional final circuit.

Position-Aware Node Pruning vs PEAP

Final position-aware node checkpoint; two retained PEAP operating points

Circuit sizes

Circuit	Abstract edges active/full	Edges closed	High-level nodes	Native PEAP token edges
Position-aware node	1,276/109,722	98.84%	100/1404 component slots	--
PEAP top-5k	4,529/109,722	95.87%	734 selected nodes	12,303.8/590,076.2
PEAP top-10k	9,369/109,722	91.46%	1115 selected nodes	27,316.3/590,076.2

Performance on the shared 500-example test set

Circuit	Accuracy	Exact	Pairwise	KL	Logit diff	Faithfulness
Position-aware node	98.8%	98.8%	100.0%	0.143	6.979	1.635
PEAP top-5k	89.2%	89.2%	99.4%	0.100	3.637	0.852
PEAP top-10k	97.4%	97.4%	99.4%	0.029	3.861	0.905

Size rule: surviving nodes imply every valid incident edge, following circuit_pruning-argo. For PEAP compatibility, Q/K/V are collapsed into one abstract connection and causal logical-position transport is included.

KL is $D_{KL}(\text{full} \parallel \text{circuit})$; logit difference is IO minus distractor; faithfulness is circuit/full logit difference.